

PIONEER JUNIOR COLLEGE

JC2 Preliminary Practical Examination
HIGHER 2

CANDIDATE
NAME

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CT
GROUP

1	6			
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INDEX
NUMBER

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CHEMISTRY

9729/04

Paper 4 Practical

28 August 2017
2 hours 30 min

Candidates answer on the Question Paper

Additional Materials: As listed in your apparatus list

READ THESE INSTRUCTIONS FIRST

Write your CT group, index number and name on all work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough workings.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Shift: _____

Lab:

For Examiner's Use	
Q1	/22
Q2	/15
Q3	/18
Total	/55

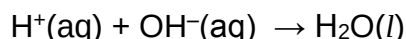
1 Experiment to investigate the behaviour of acids and bases in aqueous solution

FA 1 is 1.00 mol dm^{-3} sodium hydrogen carbonate, NaHCO_3

FA 2 is sodium hydroxide, NaOH , of concentration between 1.5 mol dm^{-3} to 2.0 mol dm^{-3}

FA 3 is 2.00 mol dm^{-3} sulfuric acid, H_2SO_4

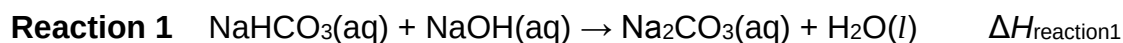
An acid-base neutralisation reaction involves reacting the two solutions, to produce water molecules. The equation for this neutralisation reaction is given below.



The reaction between $\text{H}^+(\text{aq})$ and $\text{OH}^-(\text{aq})$ is an exothermic reaction and is used to perform a thermometric titration.

This task involves **two** different acid-base reactions.

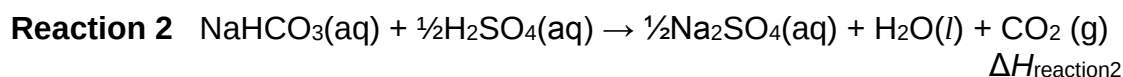
The first reaction is between **FA 1**, NaHCO_3 , and **FA 2**, NaOH .



The molar enthalpy change for **reaction 1**, $\Delta H_{\text{reaction1}}$ is the enthalpy change when 1.00 mol of NaHCO_3 reacts completely with NaOH .

You will perform a thermometric titration to determine the equivalence point of the reaction. The equivalence point is the point where the $\text{OH}^-(\text{aq})$ from the base completely neutralises the $\text{H}^+(\text{aq})$ from the acid.

The second reaction is between **FA 1**, NaHCO_3 , and **FA 3**, H_2SO_4 .



The molar enthalpy change for **reaction 2**, $\Delta H_{\text{reaction 2}}$, is the enthalpy change when 1.00 mol of NaHCO_3 reacts completely with H_2SO_4 .

In the first experiment, you will perform a thermometric titration by adding portions of **FA 2** progressively to a known volume of **FA 1**. You will continue adding **FA 2** until the equivalence point is reached and passed. Throughout the experiment you will record the temperature of the mixture after each addition of **FA 2**.

You will then analyse your results graphically in order to determine the

- volume of **FA 2** at the equivalence point, $V_{\text{equivalence}}$,
- maximum temperature change, ΔT_{max1} ,
- molar enthalpy change for the reaction, $\Delta H_{\text{reaction1}}$, for **reaction 1**.

In your second experiment, you will mix together given volumes of **FA 1** and **FA 3**.

You will then determine the

- maximum temperature change, $\Delta T_{\text{max } 2}$,
- molar enthalpy change, $\Delta H_{\text{reaction } 2}$, for **reaction 2**.

Part 1: The reaction between FA 1 and FA 2

- (a) (i) In this task you will need to record the maximum temperature of the reaction mixture when specified volumes of **FA 2** have been added. It is important that the volume of **FA 2** recorded is the **total** volume you have added up to that point when the temperature reading was made.

*Note: If you overshoot on an addition, record the **actual** total volume of **FA 2** added up to that point.*

Procedure for the reaction between FA 1 and FA 2

- 1 Fill a burette with **FA 2**.
 - 2 Place a Styrofoam cup inside a second Styrofoam cup which is held in a beaker to prevent it from tipping over.
 - 3 Using a pipette, transfer 25.0 cm³ of **FA 1** to the first Styrofoam cup.
 - 4 Stir the **FA 1** solution in the cup with the thermometer. Read and record its temperature.
 - 5 From the burette, add 2.00 cm³ of **FA 2** to the cup and stir the mixture thoroughly.
 - 6 Read and record the maximum temperature of the mixture, T , and the volume of **FA 2** added.
- Note:** If you overshoot on an addition, record the **actual** total volume of **FA 2** added up to that point.
- 7 Repeat points 5 and 6 until a total of 30.00 cm³ of **FA 2** has been added. After each addition of **FA 2**, record the maximum temperature of the mixture and the total volume of **FA 2** added up to that point in the table provided.

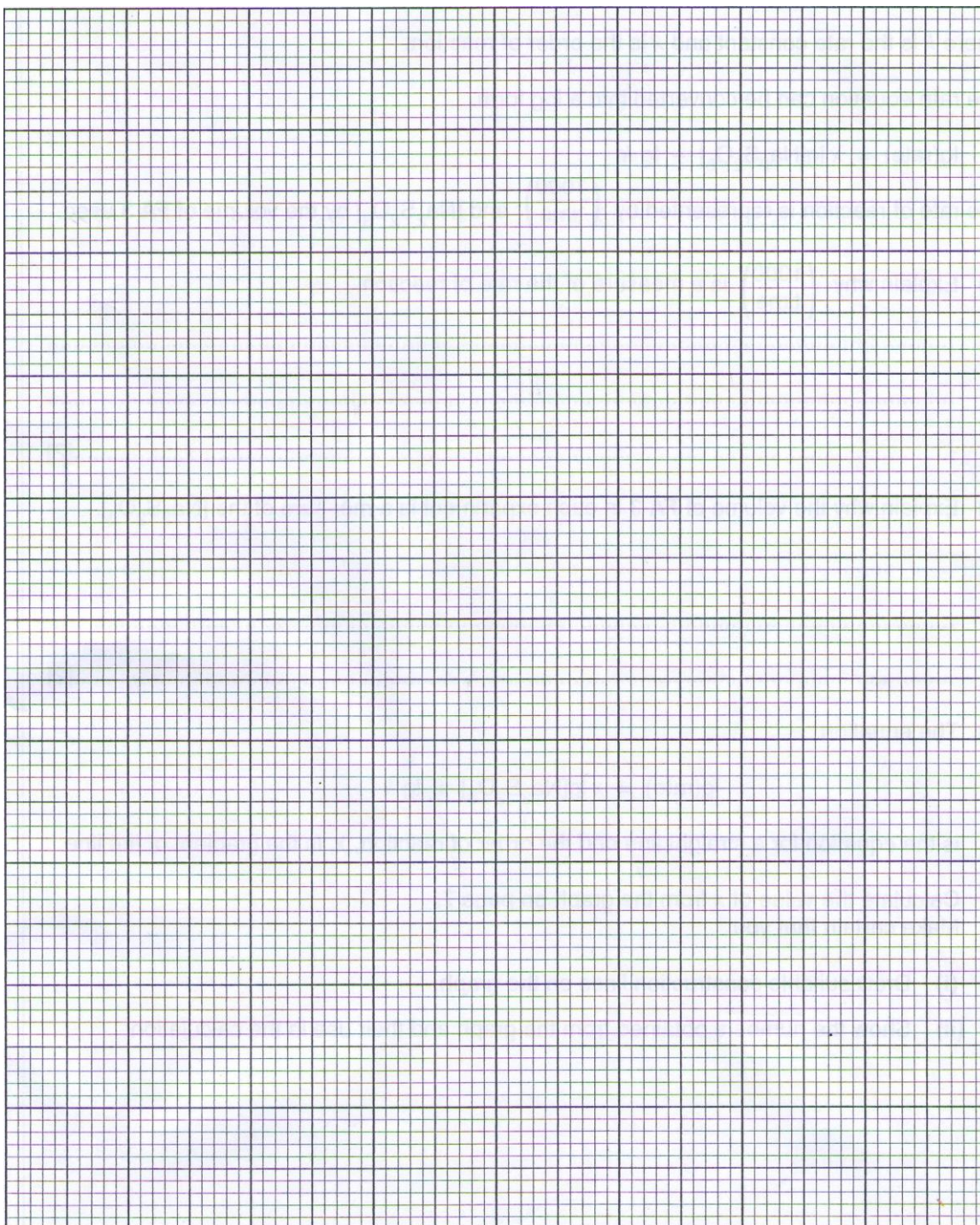
In an appropriate format in the space provided, record all values of temperature, T , to 0.1°C , and each **total** volume of **FA 2** added to 0.05 cm^3 .

Results

Initial temperature of **FA 1** : _____

[2]

- (ii) On the grid on below, plot a graph of maximum temperature, T (y-axis) against volume of **FA 2** added (x-axis).



- (iii) Draw **two** smooth lines of best fit.
- The first best-fit line should be drawn using the plotted points where the temperature is **rising**.
 - The second best-fit line should be drawn using the plotted points where the temperature is **falling**.
 - Extrapolate these lines until they cross.
- Note:** Each line should have a shape best suited to its plotted points. [3]

- (iv) Determine from your graph
- the maximum temperature reached, $T_{\max}$,
 - the maximum temperature change $\Delta T_{\max}$,
 - the volume, $V_{\text{equivalence}}$, of **FA 2** needed to completely react with 25.0 cm³ of **FA 1**.

Record these values in the spaces provided below.

maximum temperature reached, $T_{\max} = \underline{\hspace{2cm}}^{\circ}\text{C}$

maximum temperature change, $\Delta T_{\max} = \underline{\hspace{2cm}}^{\circ}\text{C}$

volume of **FA 2** used, $V_{\text{equivalence}} = \underline{\hspace{2cm}}\text{cm}^3$

[3]

Part 2: The reaction between FA 1 and FA 3

- (b) The molar enthalpy change for reaction 2, $\Delta H_{\text{reaction2}}$, is the enthalpy change when 1.00 mol of NaHCO_3 reacts **completely** with H_2SO_4 .

In this task you will calculate a value for the molar enthalpy change, $\Delta H_{\text{reaction2}}$. To do this you will need to calculate the maximum temperature change produced when 40 cm^3 of **FA 1**, NaHCO_3 and 15 cm^3 of **FA 3**, H_2SO_4 are mixed.

You should be aware that the reaction mixture will produce considerable frothing.

In this experiment, you will select a suitable apparatus to measure **FA1** and **FA 3**.

- (i) Calculate the maximum percentage errors in 40 cm^3 of **FA 1** when using
- measuring cylinder

 - burette

[1]

- (ii) Explain your choice of apparatus for measuring **FA 1**.

.....

[1]

- (iii) **Procedure for the reaction between FA 1 and FA 3**

1. Label one Styrofoam cup, **A**. Label a second Styrofoam cup, **B**.
2. Place the cup **A** inside a Styrofoam cup which is held in a beaker to prevent it from tipping over.
3. Transfer 40 cm^3 of **FA 1** into cup **A** using your chosen apparatus.
4. Stir the **FA 1** solution in cup **A** with the thermometer. Read and record its temperature, T_{FA1} . This is the initial temperature of **FA 1**.
5. Wash and dry the thermometer.

6. Transfer 15.0 cm³ of **FA 3** into cup **B**. Stir the **FA 3** solution with the thermometer. Read and record its temperature of the mixture, T_{FA3} . This is the initial temperature of **FA 3**.
7. Carefully add the contents of cup **B** to cup **A** in small portions to avoid too much frothing.
8. Place the lid on the cup and insert the thermometer through the lid. Stir the mixture.
9. Continue to stir the mixture. Measure and record the temperature, T_{mixture} that shows the **maximum** change from the initial temperature.

In an appropriate format in the space provided below, record all values of measured temperatures for this experiment.

Results

[1]

Calculations

- (c) For the purposes of these calculations, you should assume that the reaction mixture has a density of 1.00 g cm^{-3} and specific heat capacity, c , of $4.18 \text{ J g}^{-1} \text{ K}^{-1}$. Use your results from **(a)(iv)** to calculate the following.

- (i) The concentration of sodium hydroxide, $[\text{NaOH}]$, in **FA 2**.

$$[\text{NaOH}] \text{ in FA 2} = \underline{\hspace{4cm}} \quad [1]$$

- (ii) The heat change, q , for the reaction occurring in **(a)(i)**, and hence determine a value for the molar enthalpy change for **reaction 1**, $\Delta H_{\text{reaction1}}$.

$$q = \underline{\hspace{4cm}}$$

$$\Delta H_{\text{reaction1}} = \underline{\hspace{4cm}} \quad [2]$$

- (d) Use your results from **(b)** to calculate a value for the molar enthalpy change for **reaction 2**, $\Delta H_{\text{reaction2}}$.

For the experiment in **(b)**, the weighted average initial temperature, $\Delta T_{\text{average}}$, of **FA 1** and **FA 3** may be calculated using the formula given below:

$$T_{\text{average}} = \frac{(V_{\text{FA1}} \times T_{\text{FA1}}) + (V_{\text{FA3}} \times T_{\text{FA3}})}{(V_{\text{FA1}} + V_{\text{FA3}})}$$

$$q = \underline{\hspace{4cm}}$$

$$\Delta H_{\text{reaction2}} = \underline{\hspace{4cm}} \quad [5]$$

- (e) Ionic equations for neutralisation, reaction 1, and reaction 2 are shown below.

$\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$	$\Delta H_{\text{neu}} = -57.1 \text{ kJ mol}^{-1}$
$\text{HCO}_3^-(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{CO}_3^{2-}(\text{aq}) + \text{H}_2\text{O}(\text{l})$	$\Delta H_{\text{reaction1}}$
$\text{HCO}_3^-(\text{aq}) + \text{H}^+(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$	$\Delta H_{\text{reaction2}}$

Carbon dioxide reacts with solutions of carbonate ions according to the following equation.



Using your calculated answers in (c)(ii) and (d), together with the given value of enthalpy change of neutralisation, ΔH_{neu} , determine a value for the enthalpy change for this reaction, $\Delta H_{\text{reaction3}}$.

[3]

[Total: 22]

2 To investigate the kinetics of reaction in an iodine clock reaction

FA 4 is 0.100 mol dm⁻³ aqueous potassium iodide, KI.

FA 5 is 1.00 mol dm⁻³ dilute sulfuric acid, H₂SO₄.

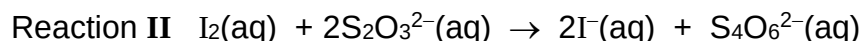
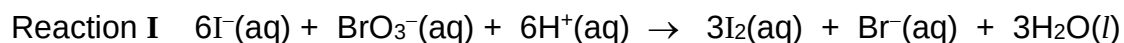
FA 6 is 0.100 mol dm⁻³ aqueous potassium bromate(V), KBrO₃

FA 7 is 5.00 × 10⁻³ mol dm⁻³ aqueous sodium thiosulfate, Na₂S₂O₃

You are also provided with starch indicator and deionised water.

This reaction starts from a solution containing potassium bromate(V), sulfuric acid, sodium thiosulfate and starch. To this, a solution of potassium iodide is added.

There are two reactions occurring in the solution. In the first slow reaction, iodine is formed.



Since there is a small but constant amount of sodium thiosulfate added to the reaction mixture, the iodine being slowly produced by Reaction I will immediately react in Reaction II until all the sodium thiosulfate has been used up. At that point, free iodine will be present in the solution, which will cause a sudden appearance of a deep blue colour as starch is present.

You will perform a series of **three** experiments, by varying the volume of **FA 6** added, with each solution being made up to the same total volume with water. For each experiment, you will note the volume of **FA 6** added, V_{FA6} , and the time taken, t , for the solution to turn blue. Using your results, determine the order of reaction with respect to BrO₃⁻.

(a) Experiment 1

- 1 Use appropriate measuring cylinders, add 10.0 cm³ of **FA 5**, 20.0 cm³ of **FA 7** and 1 cm³ of starch indicator into the conical flask labelled reaction mixture. Note: 10 drops is approximately 1 cm³.
- 2 Using a measuring cylinder, add 30.0 cm³ of **FA 6** into the conical flask.
- 3 Swirl the contents of the flask.
- 4 Measure 10 cm³ of **FA 4** using measuring cylinder. Add **FA 4** quickly into the conical flask. Start the stopwatch during this addition.
- 5 Mix the contents thoroughly by swirling the flask.

- 6 Stop the stopwatch when the blue colouration is observed.
- 7 Record the time, t , to the nearest 0.1 s.
- 8 Wash the conical flask thoroughly with water. Stand it upside down on a paper towel to drain.

Experiments 2 and 3

Carry out two more experiments by repeating steps 1 to 8, and varying the volumes of **FA 6**, so that you can determine the order of reaction with respect to BrO_3^- .

The volume of **FA 6** should not be more than that in experiment **1**. In each experiment, deionised water is added to keep the **total volume** of the final reaction mixture **constant**.

Prepare a table in the space provided to record, for each experiment:

- volumes of **FA 6**, V_{FA6} , and deionised water used,
 - all values of t , to the nearest 0.1 s,
- and**
- calculated values of $(V_{\text{FA6}}) \times t$ and $(V_{\text{FA6}})^2 \times t$ (to 3 significant figures)

*Note: As the volumes of **FA 4**, **FA5**, **FA 7** and starch indicator are constant for each experiment, you do not need to record these volumes in your table.*

Results

[3]

The rate of reaction is given by the equation:

$$\text{rate} \propto [\text{reactant}]^n$$

where n is the order of reaction with respect to the reactant investigated. The value of n can be 0, 1 or 2.

In this experiment, the rate of reaction $\propto \left(\frac{1}{\text{time}}\right)$,

and

$[\text{reactant}] \propto \text{volume of reactant used}$.

Hence, $\left(\frac{1}{\text{time}}\right) \propto (\text{volume of reactant})^n$

Rewriting, $(\text{volume of reactant})^n \times \text{time} = \text{constant}$

- (b) (i)** Explain why $\left(\frac{1}{\text{time}}\right)$ is a measure of the rate of reaction, in this experiment.

.....
 [1]

- (ii)** From your results using V_t or V_t^2 values, deduce the order of the reaction with respect to bromate(V) ions.

.....
 [1]

- (c)** In an attempt to find the order of reaction with respect to H^+ ions, a student carried out experiments **4** and **5**, by varying volume of **FA 5**, V_{FA5} .

Only the numerical values of $V_{\text{FA5}} \times t$ and $(V_{\text{FA5}})^2 \times t$ are tabulated below.

Experiment	$V_{\text{FA5}} / \text{cm}^3$	$V_{\text{FA5}} \times t$	$(V_{\text{FA5}})^2 \times t$
4	5.0	225	1130
5	15.0	75.0	1130

From the above calculated values of $V_{\text{FA5}} \times t$ and $(V_{\text{FA5}})^2 \times t$, deduce the order of reaction with respect to H^+ ions.

.....
 [1]

- (d) The order of reaction with respect to iodide ions, I^- , is given to be one.

The rate equation is written as;

$$\text{rate} = k [I^-] [BrO_3^-]^m [H^+]^n$$

where m and n are the order of reaction with respect to BrO_3^- and H^+ respectively.

In order to find half-life of the reaction, a student proposed that the rate equation can be rewritten as:

$$\text{rate} = k'[I^-], \text{ where } k' = k [BrO_3^-]^m [H^+]^n \text{ equation 1}$$

From his *equation 1*, the student then proposed that the half-life of reaction can be calculated using $\frac{\ln 2}{k'}$.

Comment on the validity of his statement.

.....

 [1]

- (e) Instead of washing and draining the conical flask as required in **step 8** of **(a)**, another student simply just poured away the reaction mixture. There was some leftover reaction mixture in the flask when he performed the subsequent experiment.

State and explain the effect on time, t , in his subsequent experiment.

.....

 [1]

(f) Plan

In **(a)** and **(c)**, the orders of reaction with respect to bromate(V) ions and H^+ were investigated using the initial rate method.

The order of reaction with respect to iodide ion was not investigated but was given as 1 in **(d)**.

Instead of doing a number of initial rate experiments, the order of reaction with respect to iodide ion, I^- , can be determined by following **one** reaction from the start to the end, in what is known as the *continuous method*.

To do so, the reacting mixture is prepared with a known concentration of I^- , and known concentrations of H^+ and BrO_3^- , both of which must be in **excess**. Over time, the amount of, and concentration of I_2 increases. The appearance of I_2 is noted by the colour of the mixture turning yellow-orange. (Due to the low concentrations of I_2 used in this experiment, I_2 appears yellow-orange)

An investigation is planned to allow you to determine the order of reaction with respect to iodide ion, I^- .

You may assume you are provided with the following.

- $0.5 \text{ mol dm}^{-3} \text{ H}_2\text{SO}_4$, $1.0 \text{ mol dm}^{-3} \text{ KI}$ and $1.0 \text{ mol dm}^{-3} \text{ KBrO}_3$,
- standard iodine solutions of various concentrations,
- deionised water

- (i)** Suggest with reasoning, the volumes of H_2SO_4 , KI , KBrO_3 and deionised water you would mix to obtain 250 cm^3 of the reaction mixture.

Reagent	Volume used/ cm^3
H_2SO_4	
KI	
KBrO_3	
Deionised water	

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.....

..... [2]

- (ii) It is possible to determine the concentration of a solution of a coloured solute using a colorimeter.

A few cm^3 of the solution is placed inside a machine, known as a colorimeter. This machine measures the amount of light that is absorbed when a specific wavelength of visible light is shone through a coloured solution. It does this by comparing the amount of light passing through the sample with the amount of light passing through the pure solvent. The amount of light absorbed is expressed as an absorbance value.

According to Beer-Lambert's Law, the higher the concentration of the coloured substance, the higher the absorbance value. Using this relationship, it is possible to determine the progress of the reaction.

A colorimeter was first used to measure the absorbance of a standard solution of iodine, I_2 , of concentration equal to that of the highest concentration of I_2 the reaction mixture can produce.

Then, the reaction mixture is prepared and placed in a 1 cm^3 cuvette. The cuvette is placed into the colorimeter and the absorbance of the solution is measured and recorded at regular time intervals.

The experiment is stopped when the absorbance reading reaches about 80% of that of the standard solution of iodine.

To determine the order of reaction with respect to iodide ion, a product time graph of absorbance against time is plotted and the order of reaction with respect to iodide ion can be determined. The colorimeter is set to use the wavelength of light that is absorbed most strongly by iodine.

Suggest a colour in the visible spectrum from which a suitable wavelength of light might be chosen. Explain your answer.

Colour:

Explanation:

[2]

- (iii) Sketch the shape of the plot of absorbance of I_2 against time graph that you expect to obtain, in the space provided below.

Explain clearly, how it can be used to determine the order of reaction with respect to iodide ion.

[3]

[Total: 15]

3 Qualitative Analysis

- (a) (i) **Before starting this analysis, ensure your Bunsen burner is turned off.**

In this question you will deduce the functional groups of two organic compounds, **FA 8** and **FA 9**.

FA 8 and **FA 9** each has only **one** functional group, and contains only carbon, hydrogen, and **one** oxygen atom.

You will perform a series of test-tube reactions and use the observations to help you deduce the functional groups present.

You are provided with liquid samples **FA 8** and **FA 9**. Perform the following tests described in Table 3.1, on **FA 8** and **FA 9**, and record your observations in the table. If it appears that no reaction has taken place, this should be clearly recorded.

Use a fresh sample of each liquid in each test.

Samples FA 8 and FA 9 are provided in sample tubes. It is essential that you replace the lid as soon as you have removed the sample for each test.

You should also prepare a fresh solution of Tollens' reagent in **Test 1** of **Table 3.1**.

One test has already been carried out and the observation is recorded in **Table 3.1**.

Preparation of Tollens' reagent

To a 3 cm depth of aqueous silver nitrate in a test-tube, add 6 drops of aqueous sodium hydroxide.

To this mixture add aqueous ammonia dropwise. Stir with a glass rod until the brown solid just dissolves.

Keep this solution for use as Tollens' reagent in the question.

Table 3.1		
Test	Observations with	
	FA 8	FA 9
1. Label a test-tube FA 8. To this test-tube, add 2 cm³ of Tollens' reagent, followed by 5 drops of FA 8. Prepare a hot water bath using hot water provided. Warm the mixture in the water bath for five minutes. Repeat using FA 9, in place of FA 8.		
2. Label a test-tube FA 9. To this test-tube, add 5 drops of FA 9 and 6 drops of aqueous sodium hydroxide. Now add iodine solution, dropwise, until a permanent yellow / orange colour is present. Warm the mixture in the water bath for two minutes.	Pale Yellow precipitate	
3. Label a test-tube FA 8. To this test-tube, add 1 cm³ of Fehling's solution A and 1 cm³ Fehling's solution B. To this test-tube, add 1 cm³ of FA 8. Warm the mixture in the water bath (at 80°C) for 15 minutes. Repeat using FA 9 in place of FA 8.		
4. Label a test-tube FA 8. Place about 2 cm depth of aqueous sodium hydroxide in this test-tube. To this test tube, add 5 drops of FA 8, followed by 5 drops of aqueous potassium manganate (VII) solution. Warm the mixture in the water bath for five minutes. Repeat using FA 9 in place of FA 8.		

[3]

		(ii)	Complete Table 3.2 below, using observations in Table 3.1 , to identify the functional groups present in FA 8 and FA 9 . In each case, give evidence to support your conclusion. [2]
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Table 3.2		
	Functional group present	Evidence
FA 8		
FA 9		

- (b) (i) State the feature(s) present in the structure in the compounds which gave a positive result for **Test 2**.

[1]

- (ii) Based on the results in **Table 3.1**, identify one of the compounds, **FA8** or **FA9**, and give its structural formula.

[1]

- (c) Solid **FA 10** contains an unknown cation. The cation is **not** NH_4^+ or Pb^{2+} .

Dissolve 1 spatula full of **FA 10** into 10 cm³ of deionised water in a test-tube. This stock solution will be required for the subsequent tests to identify the cation present.

Cation analysis

You are provided with the following bench reagents to identify the cation in **FA 10**.

- $\text{HNO}_3(\text{aq})$
- $\text{H}_2\text{SO}_4(\text{aq})$
- $\text{Ca}(\text{OH})_2(\text{aq})$
- $\text{NaOH}(\text{aq})$
- $\text{NH}_3(\text{aq})$

- (i) Based on the colour of the solution, identify the possible cations in **FA 10**.

.....
.....

[1]

- (ii) You are required to choose reagents that will allow you to carry out tests to identify the cation present in **FA 10**.

Indicate in the space below, the reagents selected and the observations of the tests. Hence, identify the cation in **FA10**.

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.....
.....
.....
.....

[4]

[End of paper]